Progress Report 4: Detecting Out-of-Context Image-Caption Misinformation with Multimodal Embeddings

DATA5500 Senior Design | Fall 2026 | Instructor: Weijie Pang | Student: Mohamed Mohamed | November 1, 2026

1. Tasks Completed

Task	What was built / implemented	Hours	Results and outcomes
Midterm presentation	Built the midterm deck (problem, dataset, pipeline, baseline and trained-model results by mismatch type, future work), cut a longer draft down to the 7-minute limit, and presented it.	4.0	Presented: baseline 0.716, logistic regression 0.751, MLP 0.757 test ROC-AUC, with the finding that both models stall near 0.76 on the 19,200-pair subset.
Full training split	Pipeline step 8: exported the provided CLIP embeddings for all 85,360 balanced pairs (71,072 / 7,024 / 7,264) into compact arrays, and mapped the same 2,400 curated test pairs into the full test split so results stay comparable.	1.5	Sanity check passed: retraining the step 5 model on the subset gives the same test ROC-AUC (0.751). Tuning now uses the full 7,024-pair validation split.
Learning curve	Logistic regression trained on 10%, 27% (subset size), 50% and 100% of the full split, with regularization re-tuned on validation at each size.	1.0	Test ROC-AUC 0.741, 0.751, 0.758, 0.763. Full data gives +0.011 (CI +0.005 to +0.018) over the subset, but the linear model is leveling off near 0.76.
Regularized neural network	Trained three MLP settings on the full split (draft settings, stronger dropout 0.5 and weight decay 0.05, and a smaller one-hidden-layer net) and kept the best on validation AUC.	2.0	Selected 2,049 to 256 to 1 (0.5M parameters). Test ROC-AUC **0.791** (CI 0.772 to 0.808), +0.028 over full-data logistic regression (CI +0.017 to +0.040) and +0.075 over the baseline. CLIP text-image rises to 0.653. Full 7,264-pair test: 0.794 (CI 0.784 to 0.804).
Calibration and error analysis	Reliability diagram and expected calibration error (ECE) for both models; listed the most confident mistakes per mismatch type with their captions.	1.5	Well calibrated (ECE 0.022). Only 87 of 2,400 test pairs (3.6%) are confident mistakes (above 80% sure and wrong), mostly person (28) and CLIP text-image (27) swaps: captions naming a specific person or place that fits the kind of photo shown.
Total		**10.0**	

Blockers, challenges, and risks

- **Risk: the 0.80 target is close but not confirmed.** Best test ROC-AUC is 0.791 (CI 0.772 to 0.808) on the curated test set and 0.794 (CI 0.784 to 0.804) on the full test split, so 0.80 is inside the confidence interval but not reached.
- **The MLP still overfits.** At its best epoch it scores 0.90 on training data vs. 0.79 on validation. More data and stronger regularization narrowed the gap, and the smaller network did best.
- **Person swaps stay hard** (0.749). The global CLIP features capture the type of scene, not who is in it, so a caption naming the wrong public figure often looks plausible.
- **The full-split embedding files are large** (1.8 GB of training pickles). Solved by exporting only the balanced pairs' vectors once (about 175 MB) and checking that the subset result (0.751) reproduces exactly.
- **Schedule:** the midterm took most of one week, so the Streamlit app has not started yet. The classifiers are ahead of the plan (Weeks 8 and 9 work is done), which covers the gap.

2. Plan for Next Week

- Finish the neural network: a small grid around the selected MLP (hidden size 128 to 384, dropout 0.4 to 0.6) and 5 random seeds, to report the mean and spread and confirm whether 0.80 is reached.
- Combine the two models: average the logistic regression and MLP scores and test the gain with the paired bootstrap.
- Start the Streamlit app skeleton: upload a photo and caption, embed both with my validated CLIP pipeline, and show the MLP's consistency score, plus preset examples from the test set.
- Prepare the optional COSMOS generalization check (Week 10): get the dataset and confirm its format fits the step 8 pipeline.
- Update the Design Document timeline to mark Weeks 7 to 9 complete.

Help needed

None at this time. Two choices guide next week's work:

- The full-split MLP (0.791) becomes the main model for the app, with logistic regression kept as the simple, interpretable comparison.
- Model work stays on frozen CLIP ViT-B/32 embeddings, as designed. Larger CLIP models would need every training image re-embedded, so they stay in future work.